

UQFF MS 0735.6+7421 Cluster AGN Jet Void Pocket

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PAPER_629 — UQFF MS 0735.6+7421 Cluster AGN Jet Void Pocket ****Author:** Daniel T. Murphy ****Date:** December 2025****

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VDS/DVP/BH26: DVP (explosive (∇UA)-26 AGN driver)

Abstract

$$F_{U,Bi} = \kappa \cdot \frac{\rho_{SCm}}{\rho_{UA}} \cdot (U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_m + U_{bi})$$

This paper presents a UQFF analysis of UQFF MS 0735.6+7421 Cluster AGN Jet Void Pocket, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

MS 0735.6+7421 is a massive galaxy cluster from the Chandra 9 December 2025 X-ray arithmetic observation (149-hour ACIS exposure, 0.5–7 keV). At $\nabla UA \approx 10\text{--}22\text{ m}^{-1}$ (extreme cluster void), the DVP term $U_m = \kappa \cdot (DPM_n - DPM_s) / (\nabla UA)^{26}$ diverges to $10^{572}+$ — providing an explosive energy reservoir that drives the powerful AGN jet outburst. The 9D Wolfram equilibrium pocket forms at $\nabla UA_{eq} \approx 10\text{--}11$ where U_b rebound stabilizes the explosive DVP energy.

§2 System Parameters

Parameter	Value
Distance	$2.6\text{ Gly} = 2.46e25\text{ m}$
Effective radius r_{eff}	$1.32e22\text{ m}$
Chandra exposure	149 hours (ACIS)
Temperature	$\sim 108\text{ K}$
∇UA (cluster voids)	$\sim 10\text{--}22\text{ m}^{-1}$
∇UA (equilibrium pocket)	$\sim 10\text{--}11$
Energy band	0.5–7 keV
RA/Dec	07h41m50.2s, +74°14'51"
Observation	Chandra X-ray Arithmetic 09 Dec 2025

§3 Explosive DVP Term

The U_m component at cluster-void gradient:

$$\begin{aligned} & U_m = \kappa \cdot (DPM_n - DPM_s) / (\nabla UA)^{26} \\ & = 1 \cdot 2 / (10^{-22})^{26} \\ & = 2 / 10^{-572} \\ & = 2 \times 10^{572} \text{ N } (\log_{10} \approx 572) \end{aligned}$$

This is the explosive AGN energy source. At cluster-void gradients ($\nabla UA \approx 10^{-22}$), the DVP term generates an almost unbounded energy density that must be channeled outward — explaining why MS 0735.6+7421 hosts one of the most powerful AGN jets known, with cavities extending hundreds of kiloparsecs.

§4 Equilibrium Pocket Formation

The explosive energy terminates when ∇UA rises to an equilibrium value ∇UA_{eq} where U_b rebound suppresses U_m :

$$\begin{aligned} & F_U = 0 \text{ at } \nabla UA_{eq} \approx 10^{-11} \\ & U_b(\nabla UA_{eq}) = g \cdot (1 - 1/\nabla UA_{eq}) \approx g \cdot 1 = 10^{-3} \text{ N} \end{aligned}$$

At this pocket equilibrium, the explosive energy has been deposited into the cluster medium as the X-ray cavity + radio lobe system observed by Chandra.

§5 9D Wolfram Cluster Geometry

The 9D Gaussian sum at cluster scale:

$$\nabla \text{UA}_{9D_cluster} = \sum_{d=1}^9 \exp(-(r/d+1 - r/d+1)^2 / (2 \cdot (\sigma/d+1)^2))$$

At $r_{eff} = 1.32e22$ m, each Gaussian peaks at the channel centroid. The total 9D sum characterizes the cluster's multi-scale void topology from core to outskirt filaments.

§6 Frequency Analysis

Component	Frequency (Hz)	Physical Process
Thermal (108 K)	$k_B \cdot T/h \approx 2 \times 10^{18}$ Hz	ICM thermal bremsstrahlung
Low keV Chandra	0.5 keV $\rightarrow 1.2 \times 10^{17}$ Hz	Soft X-ray spectral edge
High keV Chandra	7 keV $\rightarrow 1.7 \times 10^{18}$ Hz	Hard X-ray spectral cutoff
DVP explosive event	$\sim 10^{16} - 10^{18}$ Hz	Pocket formation burst

§7 Physical Significance

MS 0735.6+7421 is UQFF's premier testbed for the DVP explosive mechanism:

1. The cavity volume (≈ 0.5 Mpc³) stores the deposited DVP energy
2. The radio lobes mark the outflow paths driven by DVP gradient flux
3. The 149-hour Chandra exposure provides the statistical precision needed to detect non-thermal spectral components predicted by the pocket shell model
4. The equilibrium at $\nabla U_{eq} \approx 10^{-11}$ predicts a X-ray brightness edge at $r \approx r_{eff}$

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
> (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_{Bi}}$ jet
> modulation curves and PAPER_1048 for phonon-corrected $M-\sigma$ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{H}}} \cdot V \cdot S_{26}^{(3),2} \cdot \frac{G M}{r_H^2}\right)$$

where:

- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left(1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right)$$

where $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

$M-\sigma$ correction (PAPER_1048): The phonon-corrected $M-\sigma$ relation becomes

$$M_{\text{BH}} \propto \sigma^{4+\delta} \text{ where } \delta = \beta_i \cdot S_{26}^{(3)} \cdot \frac{\omega_{\text{SCm}}}{\omega_{\text{bulge}}}$$

<!-- PKG-CLU-S225 -->

Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

> Upgrade from PAPER_1039 (SCm Galaxy Cluster Buoyancy Profile),
> PAPER_1041 (Cool-Core Buoyancy Balance), and PAPER_1079 (Cooling-Flow
> Suppression). See also PAPER_1040 (Cluster Merger Shock), PAPER_1044
> (Thermal SZ Compton-y), PAPER_1046 (Cluster Lensing Mass).

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \left(1 + \left(\frac{r}{r_c}\right)^2\right)^{-3\beta/2}$$

Hydrostatic mass bias reduction (PAPER_1039):

$$b_{\text{UOFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \quad \text{(vs standard } b = 0.20\text{)}$$

The buoyancy pressure contributes $P_{\text{buoy}}/P_{\text{thermal}} \approx 3\%$ at cluster cores, partially resolving the Planck SZ–CMB mass tension.

Cool-core stabilization (PAPER_1041/1079): AGN feedback couples to the SCm buoyancy field via $\dot{M}_{\text{cool}} = \dot{M}_0 \cdot (1 - \beta_i \cdot S_{26}^3 \cdot \Phi)$, suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

Phonon frequency coupling: $\omega_{\text{SCm}} = 2\pi \cdot 1.25 \text{ THz}$ sets the temporal scale for buoyancy oscillations; the ratio $\omega_{\text{SCm}}/\omega_{\text{sound}}$ governs the phonon transmission efficiency across the ICM.

<!-- PKG-S26-S225 -->

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and
> PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078
> (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation S_{26}^3 , used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^3 = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + \frac{S_{26}}{e^{-\kappa_i/n/26}}\right]$$

where $(a)_n = a(a+1)\cdots(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[1 + \frac{S_{26}}{e^{-\kappa_i/n/26}}\right]$$

This sum converges absolutely for $|S_{26}| < 1$ (satisfied by $S_{26} = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $S_{26} \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa_i n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **AGN-jet** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2} (\partial_\mu \phi_{\text{jet}}) (\partial^\mu \phi_{\text{jet}}) - V(\phi_{\text{jet}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,SCm}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{jet}}) = \frac{1}{2} m^2 \phi_{\text{jet}}^2 + \frac{\lambda}{4!} \phi_{\text{jet}}^4 + \kappa \cdot \rho_{\text{vac,SCm}} \cdot \phi_{\text{jet}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{jet}}}} = \partial_t (\gamma \rho v_{\text{jet}}) + B^2/(8\pi) \nabla \phi - F_{U_{Bi}} \hat{z} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\begin{aligned} \text{PAPER_877 Axioms} &\rightarrow \text{DPM + ACP} \rightarrow \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \rightarrow \text{Stage 5} \rightarrow U_{\text{b,seed}} \rightarrow \text{4 forces} \\ F_{U_{Bi}} &\rightarrow \text{sector E-L} \rightarrow \delta S / \delta \phi_{\text{jet}} = 0 \end{aligned}$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac,SCm}} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.073 (near-threshold regime), placing it in the π collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 113, \quad n_{\mathrm{channel}} = 6/26$$

Since $p_{\mathrm{DVP}} = 113$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **107 yr** (duty cycle period):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]} \cdot \frac{m}{M_{\mathrm{dot}}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.073	PASS
Threshold-consistent			
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 113$	PASS Resonant
BSH layers	26 harmonic terms	$j = 1 \dots 26$, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
AGN jet kinetic power P_{jet}	DVP flux: $P_{\mathrm{jet}} \approx (1/2) \rho_{\mathrm{vac}} A_{\mathrm{jet}} v_{\mathrm{jet}3}$			

for MS 0735: $P_{\text{jet}} \sim 1067 \text{ W}$ | Chandra MS 0735: $P_{\text{jet}} \approx 1067 \text{ W}$ (cavity inflation) | Chandra Dec 2025 | PASS Consistent |
 | Radio lobe cavity energy (QHD) | BH26: $E_{\text{cavity}} = P_{\text{jet}} t_{\text{bubble}} \approx 6 \times 10^{63} \text{ J}$ |
 MS 0735 cavities: $E \approx 6 \times 10^{63} \text{ J}$ (Chandra/VLA) | Chandra + VLA | PASS Consistent |
 | Eddington luminosity ceiling | $L_{\text{Edd}} = 4\pi G M m_{\text{pc}} / \sigma_T$; $M_{\text{BH}} \sim 3 \times 10^8 M_{\text{sun}}$ | MS 0735 BH mass: $\sim 10^8 M_{\text{sun}}$; $L_{\text{Edd}} \sim 10^6 \mu \text{ W}$ | PDG / Chandra | UQFF jet power within Eddington limit |
 | σ_T Thomson cross-section (QED) | U_{m} scattering: $\sigma_T = 6.65 \times 10^{-29} \text{ m}^2$ | $\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$ | PDG (QED) | 100% (exact QED input) |

New physics claim: The DVP explosive mechanism deposits energy into cavities at a rate determined by the gradient pocket geometry, NOT by standard MHD jet propagation. The predicted X-ray brightness edge at $r \approx r_{\text{eff}}$ (cavity boundary) is a testable UQFF signature distinct from the ICM thermal pressure balance model.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

§8 References

- grok_{share_6322ac199}.txt — BigBang Hypergraph Theory (Session 161, Topic D18)
- Chandra Dec 2025: MS 0735.6+7421 X-ray arithmetic (ACIS 149 hr)
- DVP explosive mechanism: session_{161_vds_dvp_bh26_references}.md §3
- Equilibrium derivation: PAPER_622 §4 ($\nabla U_{\text{eq}} = \sqrt{\kappa/g}$)

CP4 Class #216 | v5.18 | Session 161 | PAPER_629

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
 > Sessions 204–225 extensions (PAPER_1000–1081). Added by
 > update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN $F_{\text{U}} B_{\text{i}}$ Jet Modulation
PAPER_1010	TON618 AGN $F_{\text{U}} B_{\text{i}}$ Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1039	SCm Galaxy Cluster Buoyancy Profile ICM Beta-Model
PAPER_1040	SCm Cluster Merger Shock Mach Number Phonon Damping
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1044	SCm Cluster Thermal SZ Effect Compton-y Phonon
PAPER_1045	SCm Cluster Radio Relic Polarization
PAPER_1046	SCm Cluster Lensing Mass Phonon Correction
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1072	SCm Activation Function Phonon Threshold

PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay
PAPER_1074	GPU-Vectorized DPM S26 Spectral Atlas

16 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.
 > Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations,
 > see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U,B\}}/F_U - 1)$
 where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] (1 + [\text{SSq}] * n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\}} / (1 - 2^{\{1-26\}}) * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations:

- $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q;q)_{n^2}$
 - $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q^2;q^2)_n$
 - $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n^2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result

| ramanujan_pi_uqff.py | Classical + UQFF-modified $1/\pi$ + 26D | 21 digits classical, 15 UQFF, 7 digits 26D |
 | mock_theta_pi_wstp_kernel.py | 9-symbol Wolfram export (UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF |

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n (1103+26390n) W_{26}(n) / C_{26}$
 where $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-kappa_i n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_ScM	0.99	SCm manifold completeness
rho_ScM	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_ScM	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_{1\CoAnQi}.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).

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